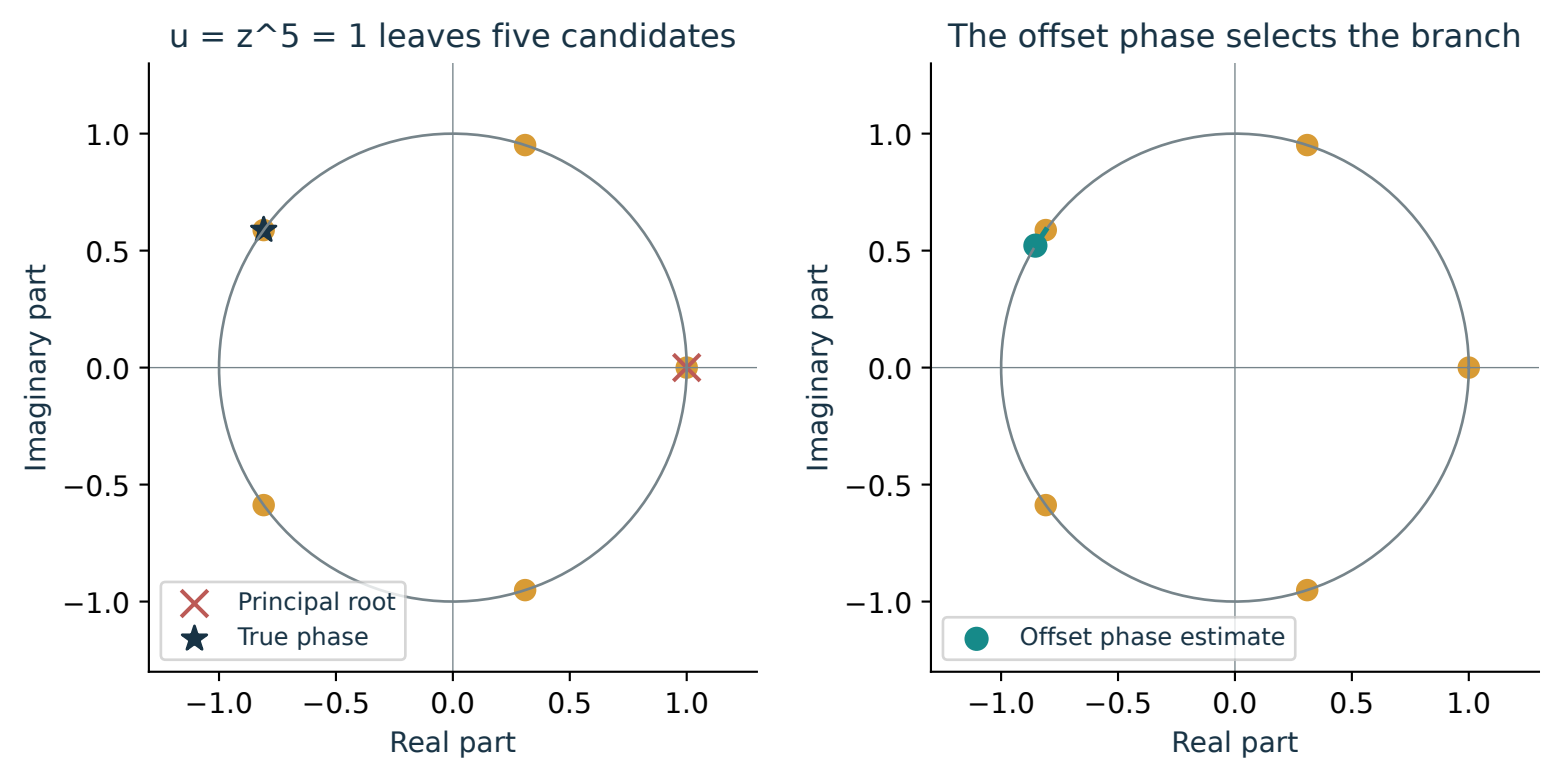
u = z^5 = 1 leaves five candidates
The offset phase selects the branch
Imaginary part
Real part
Principal root
True phase
Offset phase estimate
1.0
0.5
0.0
−0.5
−1.0